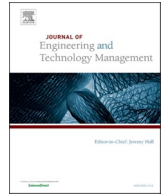




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Developing artificial intelligence (AI) capabilities for data-driven business model innovation: Roles of organizational adaptability and leadership

Swapan Ghosh

Michigan State University, United States

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ABSTRACT

More and more industrial businesses leverage AI to optimize operations and introduce new and innovative business models for competitive advantage. Businesses are collecting huge amounts of data from their business processes and plan to utilize them for better customer experience and insights. However, industrial managers are still determining the AI capabilities they need to analyze the data and develop customer-centric business models. Based on the discussions with twenty-five elite informants from five large industrial businesses, in this study, we propose a set of AI capabilities that can help managers develop data-driven business models. We offer a conceptual framework of AI capabilities and their influence on data-driven business model innovation that can guide managers in their transformation journeys.

1. Introduction

Industrial businesses are going through digital disruption, and advanced technologies such as artificial intelligence (AI), the Internet of Things (IoT), blockchain, big data, analytics, etc., can potentially transform large enterprises. According to the market research firm [Gartner \(2021\)](#), industrial businesses leverage AI solutions to develop new products, improve existing products, and extend their customer segments. AI investment in manufacturing can grow to \$9.89 billion, and AI investment in automotive can rise to \$15.9 billion by 2027 ([Fortune Business Insights, 2021](#)). There is a promise of digital servitization to change industrial businesses ([Paschou et al., 2020](#); [Sklyar et al., 2019](#); [Sjödin et al., 2021](#)), and AI technologies are one of the main drivers of such transformation ([Sjödin et al., 2021](#)). Digital servitization refers to applying digital technologies to transform a company's product-centric business into a service-centric business model ([Kowalkowski et al., 2017](#)).

However, data-driven business models (DDBM) led by AI are still evolving. The academic literature's definition of DDBM is limited. [Hartmann et al. \(2016\)](#) define DDBM "as a business model relying on data as a key resource" and identify data sources, data-related activities (visualization, analysis, etc.), customers, revenue models, and cost-saving as critical components of the DDBM framework.

Some examples of technology-led service delivery initiated by advanced technologies (AI, IoT, etc.) are GE Power's digital service delivery for power plants,¹ Siemens's digital service for intelligent buildings,² and Roll Royce's power-by-the-hour.³ However, most

E-mail address: ghoshsw1@msu.edu.

¹ GE Power digital service delivery: <https://www.ge.com/digital/customers/ge-power-transforms-service-delivery-predix-industrial-applications>.

² <https://new.siemens.com/us/en/products/buildingtechnologies/smart-buildings/digital-services.html#:~:text=Siemens%20comprehensive%20digital%20services%20portfolio,reach%20new%20levels%20of%20performance>.

³ Rolls Royce power-by-the-hour: <https://www.rolls-royce.com/media/our-stories/discover/2017/totalcare.aspx>.

current AI projects are incrementally improving standard business processes. Many AI-related studies discuss AI's value for companies, mainly manufacturing firms. However, adopting AI in conservative sectors such as manufacturing can be minimal, and its potential can appear relatively unused. The reason for that often falls into capabilities issues. AI can potentially transform a business holistically (Giacomoni, 2022; Joshi and Wade, 2020a,b); however, to achieve this goal, industrial companies must develop AI capabilities and integrate AI into their upstream and downstream value chains (Heppelmann and Porter, 2014; Sjödin et al., 2021) such that they can develop DDBM.

Researchers have studied AI capabilities in the last couple of years. However, the consensus about the components of AI capabilities could be more precise (Appendix 1). Industrial managers want to know the specific AI capabilities and how to develop them. Some researchers view AI capabilities from its model development angle (Sjödin et al., 2021), suggesting that model development and deployment capabilities are an organization's core AI capabilities. Some have taken a resource-based view of AI (Mikalef and Gupta, 2021; Chowdhury et al., 2022) and defined it as "An AI capability is a firm's ability to select, orchestrate and leverage its AI-specific resources." These resources include tangible, intangible, and human resources of an organization. Moreover, other researchers (Bawack et al., 2019; Schmid et al., 2021) view it from the technology/Information Systems angle and propose an AI capability framework consisting of AI-based sensing (computer vision, speech recognition, etc.), comprehending (text analysis, pattern recognition, etc.), acting (expert systems, robotics, etc.) and learning (machine learning, neural networks, etc.). Thus, the study has identified the first research gap related to the critical components of AI capabilities for successfully developing and implementing AI-based enterprise solutions for industrial businesses.

Researchers have also studied the impact of AI capabilities on business model innovations (Appendix 2). Business model innovation refers to "conceptualizing and implementing new business models" (Geissdoerfer et al., 2018). One stream of researchers focuses on digital servitization influenced by AI capabilities (Sjödin et al., 2021; Payne et al. 2021; Agarwal et al. 2022). Digital servitization is an emerging area for business model innovation where firms develop new digital services and improve existing ones through digital technologies, including AI (Paschou et al., 2020; Kohtamäki et al., 2019). Digital servitization is one of the strategies for value creation and capture; however, industrial businesses have limited success in this strategy. Burström et al. (2021) studied AI-led business model innovation in incumbent firms. They suggested leveraging ecosystem reconfiguration in the short term and ecosystem revitalization in the long term for optimal firm performance. Other researchers (Sena and Nocker, 2021) indicate that the new entrants in the market leverage AI capabilities, and value migration takes place from the incumbent firms to the new entrants. Though the current researchers have investigated the influence of AI capabilities on different forms of business model innovation, the study has identified the second gap in the literature: the effects of AI capabilities on the three critical components of DDBM innovation (value creation, value delivery, and value capture).

As industrial managers invest substantially in AI projects, they want to know whether AI projects can help them develop DDBMs. Thus, it raises two important research questions, and industrial managers are interested in the answers:

R1: What are the key AI capabilities for an industrial business? and

R2: How do these key capabilities influence data-driven business model innovation?

The study draws on the interviews of twenty-five elite informants who are senior managers of leading industrial companies and managing large-scale AI projects in their respective organizations. The study enhances the definition of AI capability (Mikalef and Gupta, 2021). It defines "An AI capability as the ability of a firm to leverage AI-specific resources to transform the business to a data-driven business."

Though the research on AI capabilities and their impact on DDBM innovation is limited, the current study provides two significant contributions: first, we offer an integrated data management and analysis view of AI and the capabilities needed to execute large-scale AI projects, and second, we present new conceptualizations of AI capabilities and their influence on DDBM innovation. Our conceptualization of DDBM innovation includes three critical components: AI-enhanced value creation, AI-empowered value delivery, and AI-guided value capture. Consistent with the views (Fountain et al., 2019; Ransbotham et al., 2017) that AI skills influence new business models and AI risk propensity impacts business model innovation (Reim et al., 2020; Burström et al., 2021), the study presents a series of research propositions related to AI capabilities and DDBM innovation. The study proposes a conceptual framework to help industrial managers implement their AI projects and drive DDBM innovation in their organizations.

2. Theoretical overview

2.1. AI capability of a firm

This area has limited academic research, and more information is available from practitioner journals. However, these journals have focused more on the technological capabilities of AI. Also, the gap between the expectation and execution of AI-based solutions is large in industrial businesses, and managers need to be more convinced of the transformational power of AI (Ransbotham et al., 2017). The technological capabilities of AI include personalization and profiling, predictions, pattern recognition & anomaly detection, natural language processing (NLP), object identification, and goal achievement (Ching, 2020). However, these capabilities are AI features and not organizational capabilities. According to Schmid et al. (2021), four core AI capabilities are sense, process & understand, act and communicate, and thus, AI-based solutions should support all these capabilities for business transformation. A company should develop AI capabilities in four areas (people, process, technology, and data), and interactions among these areas create six different themes (lead, learn, access, scale, automate, and secure) for success in an industrial business (Schut, 2020). The total commitment from the leadership team is essential to developing AI capabilities, implementing AI-driven business processes in the organization, and transforming the company into a data-driven company (Kruhse-Lehtonen and Hofmann, 2020).

Academic researchers (Mikalef and Gupta, 2021; Chowdhury et al., 2022) have taken a resource-based view of AI capability. They suggest that a firm can develop AI capabilities by effectively utilizing AI-specific resources. The AI capabilities are grouped into three categories: tangible (data, technology, and essential resources), human (technical and business skills), and intangible capabilities (inter-departmental coordination, organizational change, and risk propensity (Mikalef and Gupta, 2021)). Though tangible resources are available in the market, managing these resources effectively can be a competitive advantage for an organization. Data acquisition and management capability are critical for AI, and high-quality data is required (Singh, 2021; Greer, 2019; Ehi, 2018). Researchers (Johnk et al., 2021; Nguyen et al., 2019; Lokuge et al., 2018) have identified capabilities responsible for AI adoption, such as strategic alignment among top executives and customers' propensity for AI-based AI solutions, data-driven decision-making, organizational culture, and resources.

2.2. Data-driven business model (DDBM) innovation

Researchers are studying business models and their impact on firm performance for an extended period. The business model articulates an organization's value creation, delivery, and capture mechanisms (Osterwalder and Pigneur, 2010). However, Zott and Amit (2010) consider the business model an activity-based system that creates and appropriates values with its partners. At the same time, Teece (2010) suggests that the business model provides the logic and evidence for value creation and delivery. Adopting digital technologies such as AI, IoT, Big Data, etc., has accelerated business model innovation, and data has become the new fuel for innovation (Rooney, 2022; Greer, 2019). Engelbrecht et al. (2016) identify three dimensions of DDBM: data sources involved (both user and non-user data), target audience (customer-focus versus organization-focus), and technological effort (high versus low). Based on the current research, DDBM innovation can be grouped into four categories: operational efficiency (Ibarra et al., 2018; Ranta et al., 2021; Burström et al., 2021), customer centricity (Ibarra et al., 2018; Keiningham et al., 2020), ecosystem-based business models (Li et al., 2023; Ancillai et al., 2023), and pay-per-use/service-based business models (Sjödén et al., 2023; Paiola, Gebauer, 2020). Table 1 illustrates these four DDBM innovation types.

2.2.1. DDBM Innovation fostering operational efficiency

AI capabilities and other digital technologies enable DDBM innovation and can help an organization improve its operational efficiency (Fornasiero et al., 2021; Al-Surmi et al., 2021). It helps an organization innovate in its value-creation activities, such as developing new products and services (Campbell et al., 2020; Lei et al., 2022), optimizing asset utilization (Rødseth et al., 2017; Lee et al., 2019), improving supply chain visibility (Modgil et al., 2021; Silva et al., 2017), and making real-time decisions (Bek-Pedersen et al., 2019; Lv et al., 2021). It helps innovate value delivery processes by integrating with suppliers' business processes (Kenyon et al., 2022; Sjödén et al., 2021) and offering customized products/services (Sun et al., 2020; Jazdi et al., 2021). Moreover, it enhances value capture processes by optimizing business operations costs and making products and services more competitive (Ghahramani et al., 2021).

2.2.2. DDBM Innovation driving customer/partner centricity

Digital technologies (including AI capabilities) improve customer relationships and experience as companies collect and analyze customer usage data for better products and service offerings (Monod et al., 2022; Chatterjee, Chaudhuri, 2022). Though in the business-to-business (B2B) model, a company interacts with intermediaries (distributors, resellers, etc.) for product sales, more and more companies in the B2B business are directly contacting customers and innovating their services by utilizing DDBMs. For example, automobile manufacturers have developed digital marketing strategies (websites, mobile apps, etc.) to connect with customers directly. GE Appliances has signed a deal with Google Cloud for intelligent appliances and connecting with their customers now (Wood, 2021). Companies are co-developing products and services with customers leveraging AI and digital technologies (Li et al., 2021; Paschen et al., 2021). DDBM innovation improves value delivery processes as companies can segment the market from customer data using AI technologies and offer targeted products and services (Holland, 2021; Tanveer et al., 2021). DDBM-driven customer centricity can lead to a subscription-based business model. For example, GE Appliances has introduced a Smart Home subscription service for voice-activated intelligent home appliances.⁴ Thus, DDBM innovation fosters customer-centricity in industrial businesses.

2.2.3. DDBM Innovation Initiating an ecosystem-based business model

A company can develop or participate in an ecosystem-based business model powered by DDBM innovation. It can be a part of the value network, which can help the company in value creation, delivery, and capture. For example, Google Play has 3.2 million apps, and Apple App Store has 2.1 million apps (Statista, 2022). Most of these apps are from ecosystem partners utilizing Google or Apple platforms for their businesses. Google and Apple use the data to enhance their platforms and offer different services. Semiconductor and software manufacturer Qualcomm has developed "Snapdragon digital chassis," an automotive platform for telematics, connectivity, and driver assistance. Automakers can leverage this platform and create intelligent customer experiences.⁵ An ecosystem platform focal company like Salesforce leverages its AI-enabled force.com platform.⁶ It has developed a subscription-based model where independent software vendors (ISVs) can utilize the platform to build their cloud applications.

⁴ <https://www.geappliances.com/connect>

⁵ Qualcomm's Snapdragon digital chassis: <https://www.qualcomm.com/products/application/automotive>

⁶ Force.com: <https://www.salesforce.com/products/platform/products/force/?sfmc-redirect=300>

Table 1
Types of data-driven business model innovation.

Type	Value Creation	Value Delivery	Value Capture
Fostering Operational Efficiency	Optimizing cost for product development. Utilizing high-value operational assets optimally. Enhancing supply chain visibility.	Optimize product/service delivery through partnership.	Optimize costs for business operations.
Driving customer centricity	Making real-time data-driven decisions. Real-time customer information and feedback enhances product quality. Co-developing new products with customers.	Mass customization of products/services. Better market segmentation by customer data analysis using AI.	Better cost-value tradeoffs. New business models.
Leveraging ecosystem-based business models.	Real-time information sharing with partners. Developing new products/services by utilizing an ecosystem platform. Faster time to market. Ecosystem platform focal companies can develop usage-based business models.	Offering better products and services. Comprehensive solutions for customers Enhanced digital product/service delivery.	Subscription-based pay-per-use models. New business models for the focal companies. Cost savings for ecosystem partners.
Accelerating servitization	Developing smart products and services. Offering product as a service. Leveraging real-time data for service innovation	Direct interactions with customers. Proactive service issue resolution.	New business models for service business: pay-per-use, performance-based, service-based.

2.2.4. DDBM innovation accelerating digital servitization

DDBM innovation is accelerating the development of intelligent products and services. DDBM innovation drives service centricity, which in turn helps an organization in digital servitization (Kowalkowski et al., 2022). Companies are innovating their business models by offering product-service systems (PSS) business models (Kwon et al., 2019; Annarelli et al., 2019). GE Power's digital service delivery and Rolls Royce's power-by-the-hour are examples of PSS business models. It helps an organization create value by offering products as a service, delivering customized values, and capturing them through new revenue models. DDBM innovation helps manufacturers drive a service-based business (Andersen, Torben, 2016), and they have realized that digitization without servitization can lead to negative returns (Kohtamäki et al., 2019).

In summary, most AI capability research is focused on AI's technological capabilities, and some researchers focus on organizational resources and their impact on AI-based decision-making. This paper identifies the core AI capabilities and their effects on DDBM innovation by analyzing operational efficiency, customer centricity, business ecosystems, and digital servitization.

3. Research methodology

3.1. Research design and context

As AI-based business solutions and their impact on DDBM innovation are still limited in industrial businesses, a qualitative study can uncover specific ideas about DDBM innovation (Denzin and Lincoln, 2008). Within qualitative research, the study has chosen a multiple case study method to explore contemporary issues in a real-life context (Yin, 2003). A case study methodology is an excellent qualitative technique for understanding complex organizational phenomena (Eisenhardt and Graebner, 2007; Eisenhardt, 1989). The unit of analysis is a firm for this study.

The study selected five multi-national companies from five industries (industrial manufacturing, semiconductor, network equipment, life sciences, and semiconductor equipment manufacturers). These five industries are selected as they are at the forefront of adopting new technology applications in their businesses, and AI-based investments are high in these industries. We have selected five companies as they are global companies, headquartered in the USA, have annual revenue of more than \$10B, and are investing significantly in AI projects. The five significant industries with different perspectives can represent the industrial business, and a multiple case study method is appropriate to study the phenomena with real-life scenarios (Yin, 2003). This study aims to understand AI capabilities in industrial businesses and their impacts on DDBM, so it has used a theoretical sampling of AI adoptions in industrial companies (Eisenhardt and Graebner, 2007).

We conducted semi-structured interviews with twenty-five elite informants from five industrial companies: IndustryCorp (an industrial manufacturing company), SemiCorp (a semiconductor manufacturing company), NetCorp (a network equipment company), LSCorp (a life science company), and SemiEqCorp (a semiconductor equipment manufacturer). The details of the case companies are given in Appendix 3.

Managers from different business groups were selected from each company to get holistic perspectives. Altogether, twenty-five interviews were conducted. All the interview participants are middle managers or above and are responsible for driving AI solutions in their companies. For example, IndustryCorp has developed an industrial internet platform with AI capabilities, utilizing AI for asset optimization, predictive maintenance, and operations optimization solutions. They are also exploring a service-based business model for their customers. LSCorp is developing a gene-sequencing data platform with advanced AI capabilities.

3.2. Data collection

We conducted virtual interviews from January 2022 to September 2022 and studied the annual reports, analysts' reports, company blog postings, and other public documents. Before conducting an interview, we reviewed the annual reports (2018 – 2021) on digital transformation, AI, and IoT projects. It helped us understand the AI projects in those companies and facilitated our interviews. We interviewed 4–6 executives in a company with different responsibilities. The interviewees were the Director / Sr. Director of Analytics, VP of Business Operations, Director/ Sr. Director of Product Management, and Director/ Sr. Director of Business Developments (Table 2).

We interviewed one executive at a time and asked open and closed-ended questions (Appendix 4). The questions are designed to understand a company's overall AI landscape related to data collection and management capabilities, the foundation for AI development and deployment, model development capabilities, skills for AI, and ethical AI questions. To probe them further, we asked "why" and "how" questions (Newcomer et al., 2015). We conducted interviews from 60 min to 90 min and recorded those interviews. Later, we transcribed the interviews and utilized the transcriptions for data analysis. To triangulate our findings, we asked the same questions to the different executives of the same company and clarified any deviations/discrepancies. We also reviewed some of the AI analytics reports and other related documents from the company reports. For transparency and structured analysis, we created a data analysis document repository with interview recordings, transcriptions, coding, and findings of our interviews.

3.3. Data analysis

We adopted a thematic analysis approach to understand the intricacies of the complex data (Braun and Clarke, 2006). We followed a three-step process to identify the aggregate dimensions of our qualitative study (Gioia et al., 2013). We thoroughly read the interview transcripts multiple times in the first step and highlighted the keywords and phrases. In this step, we aim to understand AI capabilities and their influence on DDBM innovation. Also, we need to identify any contingency factors which can affect these relationships. We developed first-order codes using nVivo12 (qualitative analysis software) based on the discussions with elite informants. In the second step, we thoroughly analyzed the first-order codes of the AI capability, DDBM innovation, and contingency factors and developed linkages and relationships between codes. We grouped similar codes into second-order themes for AI capability, DDBM innovation, and two contingency factors (AI skillsets and Organizational adaptability). We validated these findings with the elite informants through emails and rectified any missing links and relationships (Table 2). In the third step, we developed aggregate dimensions from the second-order themes: higher-level abstractions of the AI capability, DDBM innovation, AI skillsets, and organizational adaptability. Based on these steps, we developed a conceptual data structure for our study (Fig. 1).

4. Findings

We conceptualized AI capabilities and their influence on DDBM innovation based on our analysis of the case companies. We analyzed how AI initiatives in industrial businesses accelerate DDBM innovation. In the following section, we present the AI capabilities responsible for DDBM and then conceptualize DDBM innovation. We finally offer a conceptual framework for AI capabilities and their influence on DDBM innovation based on our conceptualization.

4.1. AI capabilities of an organization

Our study has identified three interrelated AI capabilities that are influencing DDBM innovation: data capability, technological

Table 2
Elite informants from the case study firms.

	IndustryCorp	SemiCorp	NetCorp	LSCorp	SemiEqCorp
Case Overview					
Industry	Industrial Manufacturing	Semiconductor	Network Equipment	Life Sciences	Software
Size (Employee)	100,000 +	100,000 +	25,000 – 50,000	100,000 +	100,000 +
Revenues (2021)	\$50B - \$80B	\$50B - \$80B	\$25B - \$50B	\$10B - \$25B	\$25B - \$50B
Firm type	Business Unit	Parent	Parent	Business Unit	Parent
Market focus	Global	Global	Global	Global	Global
Scope of case study data					
Interviews	6	5	5	5	4
Position	Vice President, Sr. Directors, Directors	Sr. Director, Directors	Vice President Sr. Directors, Directors	Vice President, Directors	Sr. Vice President, Sr. Director, Directors
Education	MS, BS	MS, BS	MS, BS	Ph.D., MS, BS	MS, BS
Experience	20 + years	20 + years	20 + years	20 + years	20 + years
Annual Reports	2019–2021	2019–2021	2019–2021	2019–2021	2019–2021
Totals / Summary	25 in-depth interviews	400 + years of leadership experience	Education Levels: BS to PhD	Diversified industry experience	15 Annual Reports and 40 published reports examined

capability, and foundation capability. The study has also identified two contingencies (AI skillsets and organizational adaptability) influencing the relationship between AI capabilities and DDBM innovation.

4.1.1. Data capability

Industrial businesses collect factory floor / internal data from their manufacturing operations, including real-time data from machines, processes, and operating environments. Machine-to-machine interfaces, robotics automation, and statistical process control systems are standard in industrial businesses. Technological advancements, including the internet-of-things (IoT), have drastically increased data production as companies continuously collect data from various sensors, actuators, programmable logic controllers (PLCs), etc. However, most of the data lives in silos and different formats. A quote from an informant validates this point, and other informants support this view.

“Our factory is automated, and we collect data from our machines, manufacturing execution systems, recipe management systems, and other processes. However, most of the data is localized for manufacturing lines, products, or a facility, and most often, we cannot aggregate the data for meaningful analysis” SemiCorp.

Another data acquisition theme emerged from the data types in industrial businesses. Some of the data from the machines are time-series data (high volume and real-time), some are near-real-time (every hour or every four hours, etc.), and some are in batch mode (for example, data from contract manufacturers once a day). Other than data type, data structures are also different: structured (rows and columns, well-defined schema, etc.), semi-structured (extended markup languages, XML or other markup languages, key-value pair, TCP/IP packets, etc.), and unstructured (rich media, geo-spatial, audio, video, etc.). Aggregating data from different formats and frequencies is cumbersome but is needed for AI analysis, as suggested by an informant.

“We collect different data types at different frequencies. We collect machine health data continuously, and along with this, we collect environmental data and different event-related data from our business operations. These diversified data sets are necessary for comprehensive AI analysis, leading to enhanced value creation and delivery” NetCorp.

Integrating and contextualizing operational technology (OT) data with information technology (IT) data is critical for a holistic AI

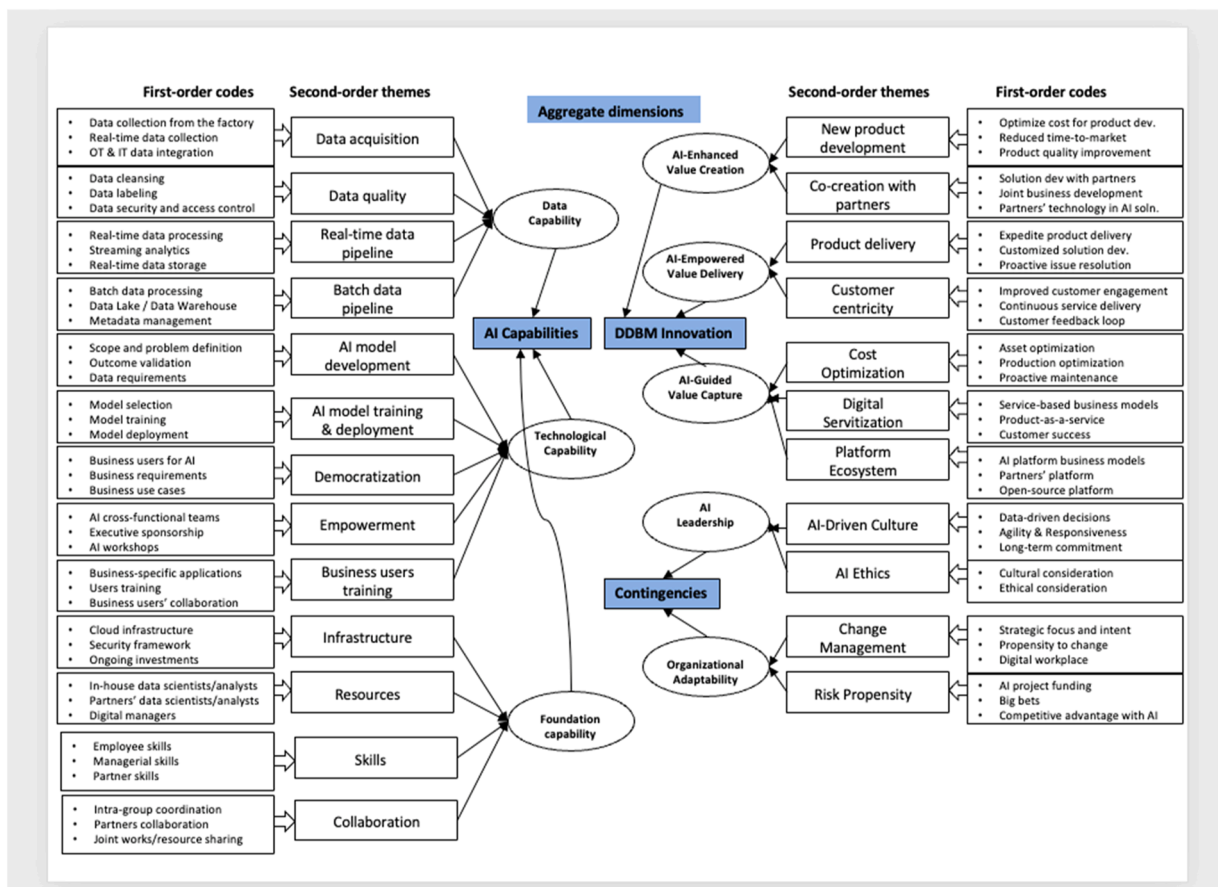


Fig. 1. Data structure.

analysis, and most of our informants expressed this view. Industrial businesses are collecting OT data from machines, sensors, and other assets; however, contextualizing those data with the IT data from enterprise systems (enterprise resource planning, manufacturing execution system, supply chain management, etc.) is challenging but necessary for AI analysis and taking proactive business decisions based on the analysis. One of our respondents voiced her opinion.

“Bridging the gap between OT and IT data is a pre-requisite for actionable AI analysis. The outcome of any AI analysis from the OT data (for example, root cause analysis of a machine problem) is to make proactive decisions and improve the operational efficiency” SemiEqCorp.

Data quality has emerged as a standard code among most interviewees. Almost everyone acknowledged that there are a lot of open-source AI algorithms and the need for better quality data to ensure meaningful AI analysis. Data cleansing is required for AI analysis as we must take care of missing values, out-of-limit data, remove duplicates, etc. Our informants strongly believe that any organization involved in AI analysis should have a good data cleansing capability. Data labeling is another crucial step for better AI analysis. An organization should have good data labeling, annotation, and classification capabilities (including auto labeling). A proper data security framework is critical for AI analysis because a data set should be protected from unauthorized access, tampering, deletion, etc.

All our informants expressed the importance of data security and governance. A quote from one of our informants clarifies this point.

“As we move towards a data-driven business, data security is paramount. We must protect our AI data sets from internal/external threats and unauthorized access to safeguard our organizations from loss of reputation, financial setbacks, and brand erosions” LSCorp.

Most of our informants expressed the need for a real-time data pipeline framework. As industrial businesses continuously collect OT data from the factory floors/internal operations, the data should be streamed in real-time, analyzed on a stream, and then stored in different repositories. For example, machine time-series data can be stored in a NoSQL database, an Object Store, or a Big Data store for analysis. As one of our informants expressed, an organization should have a real-time data pipeline capability for AI analysis.

“We collect real-time time-series data from multiple machines using sensors; we like to stream the data continuously and developed a real-time data pipeline using the open-source Apache Kafka framework. We are developing real-time AI algorithms to run against the streaming data for timely analysis to enhance our value creation and delivery processes” SemiEqCorp.

Industrial businesses have been integrating data from enterprise systems (ERP, CRM, SCM, etc.) for decades and storing these data in data warehouses and data lakes. Though organizations are using ETL tools for such purposes, in recent years, batch data pipelines have gained importance as industrial businesses are developing data lakehouses to store and analyze real-time and batch data for AI-based analysis. A good batch data pipeline capability can help an organization contextualize OT and IT data and enhance its value creation and delivery activities. A quote from one of our informants indicates such a theme.

“Traditionally, batch data integration is a common capability in our company as we have developed various ETL routines to store the data in a data warehouse. However, we are developing a data lakehouse to facilitate better AI analysis and data-driven business models.” NetCorp.

In summary, an organization's data capability is significant for organization-wide AI analysis. It comprises data integration, quality management, and pipeline development capabilities. Data capability can be a significant competitive advantage for an organization.

4.1.2. Technological capability

AI model development is critical for AI-based analysis. Though open-source AI algorithm libraries are standard and freely available, algorithms alone can not be used for meaningful AI analysis. Strong business knowledge and expertise are required for such research. The AI model's scope and the business goal should be defined and agreed upon by the managers. It can include the objectives of the AI project, the timeline, and the project's expected outcome. Though some projects may be exploratory, they should have a scope and timeline for such experiments. The managers should have some ideas about the success factors of such AI projects. Some of our informants voiced that they still have some reservations about the outcomes of the AI projects and the expected return on investment (ROI) from such endeavors.

“The scope and expected outcomes of the AI projects should be defined clearly. We are still skeptical about the success of our AI projects and their impact on our businesses. However, we realize the importance of AI-based decision-making in our company” IndustryCorp.

The test data requirements should be finalized once the scope and outcome are defined. Most of our informants expressed the importance of good test data sets. The data scientists interact with business groups and domain experts to develop good test data sets for AI models. For example, the root cause analysis of a product quality issue can be an AI project with the scope to analyze the product quality for a certain period and identify probable root causes for quality deviation. The outcome of such analysis can show process problems, and we can monitor the process parameters and take corrective actions as needed. The lack of good quality test data can seriously impact AI model development, and companies are developing good test data libraries for AI model development. One of our informants from LSCorp explained the role of AI test data sets.

“The success of AI models depends on the test data sets, so we take this task seriously. Our data scientists and engineers collaborate with business leaders, process engineers, and product engineers to understand the process/product-related issues and select test data sets from the machine/process logs, sensors, etc. We have developed test data set libraries for ongoing AI projects” LSCorp.

Another crucial step in developing the AI model is training and validation. For a complex AI model (such as anomaly detection during semiconductor manufacturing), the dataset should be split into a training, validation, and test dataset. A training dataset should include multiple parameters and business scenarios so the model can learn from the training dataset. For example, if we are modeling a neural network, the training dataset should cover all possible combinations of business issues. Once the model is trained, the model must be validated against the dataset. An extensive validation dataset is required for proper model performance if several hyperparameters exist. The companies use AutoML features to validate the AI model and reconfigure the hyperparameters for better performance. After validation, the model should be tested against a test dataset for performance. This model training and validation are critical capabilities that impact model performance and decision-making based on these AI models. One of our informants suggested the importance of model training and validation and how they interact with equipment manufacturers to develop suitable datasets.

"AI model training and validation are critical for AI models and data-driven decision-making based on these models. So, we work with our business partners and equipment manufacturers to develop good datasets for AI model training and validation." SemiCorp.

AI democratization is a novel concept, and large technology cloud companies (Google, Microsoft, Amazon Web Services, etc.) are driving this concept. In our study, some of our case companies have started AI democratization. However, the scope of such democratization efforts is limited. In AI democratization, the power of AI analysis moves from experienced data scientists to everyday business users who can run AI models with their business data for meaningful data-driven business decisions. Business intelligence (BI)/analytics democratization has progressed well within industrial businesses due to the availability of BI tools. Business users can generate daily reports and analytics from their business data. AI democratization is still evolving as easy-to-use tools are being introduced and gaining momentum. One of our respondents expressed her views.

"Most of our AI models are developed and managed by our data scientist teams. However, we are deploying some AI models where business users can run the AI models against the deployed AI models. Though this is limited to one product, we are gradually deploying more such models for the business users." IndustryCorp.

AI marketplaces are slowly evolving where AI data scientists can upload their data and algorithms, and other businesses can use those. Kaggle (acquired by Google) is an AI algorithms and datasets marketplace. Several startup companies are building such AI platforms. Industrial companies are developing their own AI internal libraries (ML notebooks), and business users from different groups can utilize those models and datasets. One of our respondents expressed similar views.

"We are developing common ML notebooks for a specific business with associated datasets. We encourage other businesses to leverage that for their AI analysis with their datasets. This is our effort for AI democratization." NetCorp.

Thus, AI technological capability is essential for large-scale AI projects. Companies can have in-house AI model development and testing resources (data scientists and analysts) or augment those resources from their partners. Getting large amounts of business-specific data for model testing is essential, and companies should have domain experts to facilitate the data collection and testing processes. Companies should develop proper training and empowerment programs such that business users can leverage those models in their regular business functions. Still, data scientists are managing these activities, but AI analysis should be democratized for enterprise-wide adoption.

4.1.3. AI foundation capability

Proper infrastructure is a prerequisite for AI projects, and a company should develop the proper AI infrastructure for AI-powered data-driven business models. More and more industrial businesses are driving AI initiatives to gain a competitive advantage and develop proper AI infrastructure for such deployments.

Cloud infrastructure is crucial and includes computing environments, storage requirements, networking requirements, and a proper security framework for AI analysis. Developing and managing cloud infrastructure is critical for AI projects. All case companies have developed cloud-based AI solutions and utilize cloud vendors' cloud infrastructure (public cloud). The cost of running AI algorithms on the cloud is a concern, and the case companies are managing their workloads for optimal price. Storage is another component; in our case, companies manage different storage configurations (raw data storage, big data stores, data warehouses, data lakes, etc.) for their workloads.

A low-latency network and high storage speed are required for real-time data analysis. Robust security infrastructure should support all the technology infrastructure for data security, privacy, and compliance. An informant from SemiCorp articulated their technology infrastructure.

"We work in a hybrid environment, with on-premises technology infrastructure with proper CPUs/GPUs for product simulations, root cause analysis, anomaly detections, etc.; we also leverage cloud vendor's infrastructure for ongoing AI projects. We have a well-defined security framework for AI analysis and review/validate our cloud partner's security infrastructure." SemiCorp.

Besides a suitable infrastructure, AI resource capability is another critical component of AI infrastructure. All case companies have their own data scientists and data analysts. However, they also expressed a concern that most of their data scientists need more domain/business knowledge about their operations. So, data scientists and analysts work with business leaders, product engineers, and process engineers to model business issues. Case companies are also training these data scientists in different business segments. All case companies have strategic partnerships with software services companies to augment their AI resources with partners' skilled data scientists and analysts. An informant from IndustryCorp expressed such views:

"We have data scientists and analysts for various AI projects. We train our data scientists in our businesses so that they can develop proper AI models. However, from time to time, we include partners' resources to augment our teams. We are building in-house AI capabilities within our teams." IndustryCorp.

AI skillset is an essential component of AI-based analysis. AI knowledge and collaboration skills are critical for AI projects and developing a data-driven business model. Employees should possess proper AI skills, including proficiency in programming languages (Java, Python, R, C++, etc.), statistics & applied mathematics, distributed computing, and familiarity with ML libraries (TensorFlow, Spark MLlib, Scikit-learn, etc.). Our case companies have data scientists and AI engineers with these skills. However, all our informants agreed that there is a need for more AI engineers with sound business and programming knowledge, impacting AI adoption in their industries. One of our informants from *SemiEqCorp* told us about this:

"AI adoption in the industry is greatly impacted by the shortage of skilled AI engineers with proper programming and mathematics knowledge. We have recently slowed down our AI initiatives and have organized in-house training of our existing employees." SemiEqCorp.

Strategic partners are helping case companies implement large-scale AI projects. Tool vendors also support the case companies and organize AI boot camps for their customers. The managers involved in the AI projects should be familiar with AI tools and models, which could help them implement large-scale AI projects.

Other than AI knowledge, collaboration skills are crucial for AI projects. Companies should develop inter-group and intra-group collaboration to define the AI projects' scope and outcome. Discussions across different lines of business are required for test data preparation, model validation, and, finally, model deployments into business processes. Collaboration with strategic partners and tool vendors is essential for AI projects. Most of our informants agree regarding the importance of collaboration skill, and one of our informants from *IndustryCorp* expressed his views:

"Inter and intra-group collaboration with stakeholders greatly impacts our AI capabilities. Since AI projects span multiple business units, all of us must be in sync and support the AI initiatives. We also collaborate with our partners and plan our joint resource allocation." IndustryCorp.

4.2. AI-powered DDBM innovation

AI influences DDBM innovation by enhancing value creation, delivery, and capture processes. This study has identified these three processes: AI-enhanced value creation, AI-empowered value delivery, and AI-guided value capture. Industrial businesses can leverage AI to innovate their DDBMs.

4.2.1. AI-enhanced value creation

New product development is a crucial differentiator for an industrial business due to the shrinkage of the product life cycle. Industrial managers are challenged to design and develop a product faster at optimal costs and bring it to the market before their competitors (*DDBM innovation type: Fostering operational efficiency*). Industrial businesses are utilizing AI-augmented digital twins to create a digital replica of a physical component or a process, and the twins are helping them in their NPD processes. A comprehensive data capability can influence digital twins' development, testing, and deployment. A company should have developed proper data pipelines to simulate the physical environments, and a cloud-based scalable infrastructure is required for such deployments. One of our informants expressed the utility of digital twins for NPD.

"We have created a digital twin for a machine, which utilizes a massive amount of design, manufacture, inspection, repair, online sensor, and operational data. It leverages a physics-based model, AI, and advanced analytics to forecast the health and performance of the machine. It has reduced the NPD time significantly." IndustryCorp.

AI capability also transforms legacy product life cycle management (PLM) systems into a data-driven, cloud-based PLM solution by interconnecting and sharing data from different product silos such as computer-aided design (CAD), engineering, production, and supplier systems. Such integration increases the visibility of the enterprise supply chain (*DDBM innovation type: Fostering operational efficiency*). A company should have accurate data and technology capability to integrate and contextualize data from different silos and proper cloud infrastructure to execute that. The adoption of cloud PLM systems has significantly reduced the NPD cycle time. Some of our informants agreed with this view. One informant told his thoughts:

"Cloud PLM system has improved the NPD processes, and we can design our products faster as we share information from different stakeholders in real-time. We can identify any bottlenecks sooner and can rectify them quickly." NetCorp.

AI capabilities have significantly improved product quality. Our case companies collect product quality data during manufacturing processes and utilize AI algorithms such as anomaly detection to discover process or product problems during manufacturing. A proper data pipeline to integrate real-time and batch data is required for quality management. It has significantly improved product quality and reduced product development costs (*DDBM innovation type: Fostering operational efficiency*).

"Online statistical process control systems are in place for a longer period; however, now, we can integrate, contextualize and analyze the data in real-time with AI analytics and take necessary corrective actions as needed." SemiEqCorp.

AI and advanced technologies also improve value co-creation with strategic partners. Our informants believe a company cannot develop all digital components for a customer and often rely on a partner's technology for specific value delivery. For example, AI-based edge analytics is essential to a real-time asset management system. One of our case companies (IndustryCorp) has co-developed an asset performance management system by integrating an edge analytics component from a strategic partner. The companies should develop data and technology management plans for value co-creation (*DDBM innovation type: Driving customer/partner centricity*).

Thus, we propose:

P1-1: AI data capability positively influences AI-enhanced value creation. An industrial business with strong data management capabilities can develop new business models for value creation.

P1-2: AI technological capability positively influences AI-enhanced value creation. An industrial business with strong AI model development and deployment capability can enhance its value-creation activities.

P1-3: AI foundational capability positively influences AI-enhanced value creation. It is a fundamental capability for technology-driven value creation.

4.2.2. AI-empowered value delivery

AI capabilities are accelerating product delivery. Our case companies are managing and monitoring their assets (e.g., machines) continuously at their client locations and leveraging AI algorithms to understand asset problems proactively and rectify them before any problem occurs (*DDBM innovation type: Fostering operational efficiency*). Proper data and technological capability are required to collect and integrate asset data from various destinations. A company should have a good foundation capability to manage these data sets needed for analysis. AI capabilities are also helping our case companies to customize the product according to the requirements of their clients (*DDBM innovation type: Fostering operational efficiency*). One of our informants expressed her thoughts:

"AI capabilities enable us to monitor and manage high-value assets at our customer locations continuously, and we engage in proactive maintenance activities to resolve any issues expeditiously. Real-time data from our customers is helping us customize our products for individual customers," LSCorp.

Some of our case companies leverage AI and enhance the supply chain solution for better product delivery with near-real-time suppliers' information to manage the supply chain effectively during any disruption (*DDBM innovation type: Fostering operational efficiency*). The COVID-19 pandemic disrupted the supply chain, and companies are developing AI-powered supply chain strategies to mitigate this risk. To create a near real-time supply chain visibility solution, a company must have valid data and technological and foundational capability to manage supply chain data from different suppliers. One of our case companies is developing a zero-latency supply chain solution.

"We have developed a near zero-latency supply chain orchestration framework where we receive suppliers' events on a real-time/ near real-time basis and analyze the impact of any suppliers' disruptions using AI algorithms. We take proactive actions to meet our customer delivery commitments" NetCorp.

AI capabilities are helping industrial businesses to develop data-driven customer engagement. Our case companies have improved their customer relationships with proactive issue resolutions. Real-time monitoring of customers' assets and participation in customer improvement programs have contributed to this success (*DDBM innovation type: Driving customer/partner centricity*). One of our informants expressed her views:

"We actively participate in our customer improvement programs and take ownership of solving customer problems, thus improving our relationship from a vendor to a strategic partner. Customer success is important for our success" LSCorp.

Cloud-based digital platforms augmented with AI help industrial businesses deliver continuous customer service (*DDBM innovation type: Initiating ecosystem-based business models*). Our case companies continuously enhance their digital offerings and incorporate easy-to-use AI solutions to improve their customers' productivity. More and more customers are adopting public cloud solutions, and our case companies continuously enhance those solutions. According to a director of the supply chain in SemiEqCorp:

"More and more customers are embracing cloud-based business application platforms maintained and managed by our team. Our customers can use the latest solutions as and when they are interested. The continuous service delivery model helps our customers to select the best solutions for their businesses" SemiEqCorp.

Based on this discussion, we propose:

P2-1: AI data capability positively influences AI-empowered value delivery, and a robust data management capability can help an industrial business to provide a service-based business model.

P2-2: AI technological capability positively influences AI-empowered value delivery. Strong AI technological capability can open different value delivery paths for an industrial business.

P2-3: AI foundational capability positively influences AI-empowered value delivery, and with a strong foundation, an industrial business can implement multiple delivery channels.

4.2.3. AI-guided value capture

AI capabilities are transforming industrial businesses' value-capture processes. Industrial managers utilize AI and advanced technologies for cost optimization, digital servitization, and platform ecosystem-based business models. Industrial businesses must

have integrated data and technological capabilities to develop and manage high-scale IoT and AI-based solutions.

Industrial businesses optimize their costs by leveraging AI and advanced technologies to monitor high-value assets in real-time, collect data continuously, and analyze it for managerial decision-making (*DDBM innovation type: Fostering operational efficiency*). They must have proper foundational capabilities, including infrastructure and AI skills, to implement these solutions. Most of our informants have implemented some form of IoT/AI-based solutions for asset performance management, which in turn help them maintain their assets proactively and reduce the cost of their operations while improving productivity. The Sr. Director from NetCorp expressed his thoughts:

“Asset performance management for our high-value assets is a priority, and we have implemented IoT and AI-based applications for data-driven decision-making. We have reduced our maintenance costs and improved the asset performance, and we can meet customer demands in a predictable way” NetCorp.

AI capabilities are helping industrial managers to monitor factory production on a real-time basis. Earlier, managers could not consolidate production information from multiple lines and react proactively to any deviation from the targets. Most of our case companies have developed executive dashboards using AI, IoT, and other advanced technologies to monitor their real-time factory performances (*DDBM innovation type: Fostering operational efficiency*). According to one of our informants:

“We have developed a real-time executive dashboard and monitor key performance indicators of our business. By leveraging AI and advanced technologies, we continuously monitor our business performances, and our managers make proactive decisions to fulfill our customer commitments.” LSCorp.

AI and advanced technologies have accelerated digital servitization in industrial businesses. More and more companies are augmenting their service-based businesses and experimenting with product-as-a-service business models (*DDBM innovation type: Accelerating digital servitization*). One of our case companies is monitoring assets from power companies and analyzing the petabytes of data generated from these assets. They have a team of service technicians who are notified instantly of any maintenance issues. It has improved the power asset performance and reduced annual maintenance costs significantly. They are experimenting with subscription-based business models for power customers. Though most informants have expressed their desires for a subscription-based business, they articulated that the operating models should be changed before they can implement service-based business models (*DDBM innovation type: Accelerating digital servitization*).

Some of our case companies are developing AI-based digital platforms and inviting their ecosystem partners to build business solutions leveraging their platforms (*DDBM innovation type: Initiating ecosystem-based business models*). For example, one of our respondents articulated their endeavor in this area:

“We have developed a smart connected digital platform for city services with a subscription-based submission model. Our partners can develop smart city applications utilizing our platform. We provide device management and security, and subscribers are authorized to access and have different subscription models based on the platform” NetCorp.

Based on this discussion, we propose:

P3-1: AI data capability positively influences AI-guided value capture, and without a strong AI data management capability, an industrial business cannot demonstrate its' value capture advantages against its competitors.

P3-2: AI technological capability positively influences AI-guided value capture, and an industrial business with differentiated AI technological capability can create a competitive advantage against its competitors.

P3-3: AI foundational capability positively influences AI-guided value capture. A strong foundation capability can significantly improve value-capture processes for industrial businesses.

4.3. Contingencies affecting AI capabilities and DDBM innovation

This study emphasized two contingencies that could affect the relationship between AI capabilities and DDBM innovation. Since AI-based business model innovation is a resource-intensive enterprise initiative, it requires alignment and support from the executive leadership team and middle managers. A bottom-up approach to such an initiative is seldom successful in an organization. Another factor, organizational adaptability, could influence such a relationship because managers and other employees should adapt to new ways of doing business.

4.3.1. AI leadership

AI-driven culture, such as data-driven decisions, long-term commitment of the managers, and agility and responsiveness, can affect the development of AI capabilities and business model innovation. The organizational culture can influence an organization's data, technology, and foundation capabilities such that AI-based decision-making is adopted throughout the organization. Employees at any level can use AI algorithms to augment their decisions. Though intuition and domain knowledge are essential for business decisions, AI can inject more objectivity into the decision-making processes. It can automate the decision workflows with possible options and facilitate value creation, delivery, and capture. AI leaders can cultivate entrepreneurial mindsets and motivate employees to experiment with new ideas and think outside the box, enhancing an organization's competitiveness. One of our informants from *SemiEqCorp* expressed such views:

“AI leaders can act as a transformation agent and help implement organization-wide AI solutions such that employees can augment their decisions with AI model recommendations, which can help significantly impact our overall performance.” SemiEqCorp.

AI ethics is becoming a critical issue, and managers must know the risks associated with AI-based applications. It can directly impact an industrial business's AI capability and data-driven business decisions. Managers should develop and promote AI ethical programs in their organizations. AI leadership is responsible for implementing ethical AI practices across the organization and the top leaders should buy into these programs and support them. The AI risk committee should vet the application of AI in specific departments, such as human resources (HR), for recruitment purposes. The data scientists and analysts developing AI models should know AI ethics, and the executive team should understand AI-based decision-making. One of our informants from NetCorp told us,

“We take AI ethics very seriously. We have an AI risk assessment group, and they analyze our company's AI-based applications. They interact with the data scientists and model developers to reduce/eliminate any biases towards training and deployment of the model.” NetCorp. Other informants also expressed similar views.

Thus, the intervention of AI leadership accelerates (moderation effect) the implementation of AI capabilities and their influence on business model innovation.

Proposition 4. *AI leadership is a crucial contingency (moderator) of AI capability development. (a) Better leadership positively impacts AI capability, but (b) the realized impact depends on the organization's ability to manage the DDBM innovation ethically.*

4.3.2. Organizational adaptability

In a volatile and dynamic environment, an organization should be adaptable to developing DDBMs by leveraging its AI capability. Though developing proper AI capability is required, an organization must ensure that the AI project implementations go smoothly and that appropriate change-management processes are in place to manage AI-based solutions and DDBMs. The impact of new technologies on different parts of businesses should be monitored, and corrective actions should be taken. Another area for adaptability is risk propensity, as AI projects may have higher risks, and a quick return may not be possible. Thus, change management and risk mitigation influence AI capability development and data-driven decisions.

Management's strategic intent is essential for the long-term success of an AI solution. Senior managers and executives should be involved in supporting a high-value AI-based business solution. There should be buy-ins from the stakeholders representing different business groups within the organization. Our informants agreed that propensity to change and strategic intent are critical factors in developing DDBM. An informant from NetCorp articulated it as follows:

“Our senior managers are involved in our business-critical AI projects, and we obtain the buy-ins from different business leaders during the scope and outcome definition of such projects. We have total support from our management team” NetCorp.

Risk propensity is another theme that came out from our discussions. AI projects may have a higher risk of failure, and managers should be willing to take risks and experiment with AI-based DDBMs. Some managers are risk-averse and tend to maintain the status quo. Senior managers should support AI projects and encourage managers to take calculated risks. There should be proper funding for AI projects, and the managers should be willing to take big bets. Most of our informants agreed with this theme, and an informant from LSCorp said:

“We realize the inherent risk of AI projects and AI-based DDBM. However, we have taken measured risks to compete in the market as our competitors have introduced similar products. Our senior managers approve our initiatives and allocate appropriate funds for such projects” LSCorp.

Thus, organizational adaptability (for managing higher risks of AI projects, strategic focus and intent, and proper change management practices) accelerates (moderation effect) the implementation of AI capabilities and their influence on business model innovation.

Thus, we propose:

Proposition 5. *Organization adaptability is an essential contingency (moderator) of AI capability development. (a) Higher organizational adaptability positively impacts AI capability, but (b) the realized impact depends on the organization's willingness to manage the DDBM innovation.*

5. Discussion and conclusion

Although researchers have been focusing on AI capabilities for the last couple of years (Mikalef and Gupta, 2021; Chowdhury et al., 2022; Sjödin et al., 2021; Payne et al., 2021; Agarwal et al., 2022), an extensive view of AI capabilities is missing. This study has attempted to provide a comprehensive picture of AI capabilities, including data, technological, and foundational capabilities. Thus, this study has extended the influence of AI capability and given a new definition. Based on the business model theory, our interviews with senior executives of Fortune 100 technology companies, data structure, and propositions, we present an integrative framework of AI capabilities and their influence on DDBM innovation (Fig. 2).

Data acquisition is a critical capability that influences new product development, customer centricity, cost optimization, and digital servitization of industrial businesses. Integrating operational technology (engineering/factory floor/operational - OT) data with information technology (Enterprise resource planning, Supply chain management, etc. - IT) data is a challenge for all respondents;

however, based on our discussions, life sciences and network equipment manufacturers have developed efficient data pipelines to integrate OT and IT data. Industrial manufacturing has more siloed business processes as their engineering and enterprise business systems are not fully integrated.

Technological capabilities for AI model development, deployment, and training are another critical capability that influences AI-driven value creation, delivery, and capture processes. All the companies in our research have developed and deployed several AI models for asset optimization, preventive maintenance, customer sentiment analysis, product design, and different quality initiatives. Based on our discussion, semiconductor and life sciences managers are more receptive to deploying AI models in business-critical functions. In contrast, industrial manufacturing and semiconductor equipment manufacturers are developing and deploying AI models at a slower pace. There are significant hurdles in model democratization and business user training as almost all companies do not have sufficient data scientists and trained business analysts to work with the business users. All our respondents have developed plans to mitigate these challenges.

Foundation capabilities such as cloud-based infrastructure, employee skill sets, and resources, including external partners, are other critical AI capabilities that influence AI-driven value creation, delivery, and capture processes. All respondents agreed that they have suitable cloud-based infrastructure for AI development and deployment. However, they are very worried about the cost of such infrastructure and the lack of adequate return on investments (ROIs) for such deployments. Industrial manufacturing and semiconductor equipment managers expressed more infrastructure and resource requirements for AI deployments. The lack of skilled resources is a concern for all respondents, and most of them are mitigating this risk by partnering with 3rd party IT consulting companies for short-term AI resources. The managers collaborate with their infrastructure partners (Amazon Web Service/AWS, Google, Oracle, etc.) for infrastructure management and AI model development.

The current study is based on five global industrial companies representing five different industry segments; hence, the findings apply to any large industrial company with global operations. The study advances the knowledge of AI capability, DDBM innovation, and their interactions. Based on this integrative framework, we offer several contributions to the body of knowledge of AI capability and DDBM innovation. First, we extend the definition of AI capabilities and include critical components of AI-based business application development, such as data, technology, and foundation capabilities for enterprise AI solutions. Earlier researchers have defined AI capabilities with a limited scope and focused on AI model development (Sjödin et al., 2021), a resource-based view (Mikalef and Gupta, 2021; Chowdhury et al., 2022), and AI adoptions (Johnk et al., 2021; Nguyen et al., 2019; Lokuge et al., 2018)) as the AI capabilities of an organization; however, we argue that systematic good quality data collection and integration, proper AI model

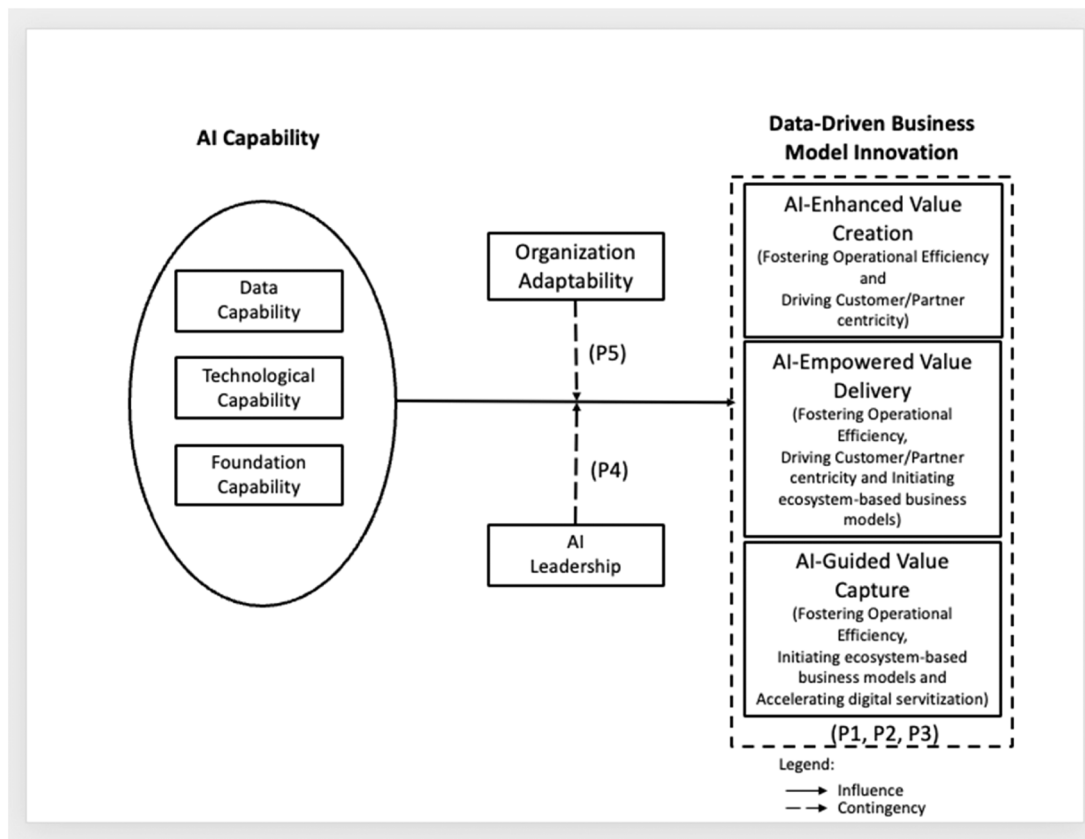


Fig. 2. Conceptual framework of data-driven business model innovation.

development and deployment and secured cloud-based analytics infrastructure are core AI capabilities. Second, we advance the theory of business model innovation and how DDBMs are changing the existing business models, such as AI-enabled value creation, AI-empowered value delivery, and AI-guided value capture. We position that an industrial business with strong AI capabilities can develop better values for their customers by developing new products and services faster and by creating industry-specific solutions with their ecosystem partners, delivering better values by optimizing their high-value assets and fulfilling customer needs proactively and by deploying service-based pay-per-use business models for their customers. Third, we propose two contingencies (AI leadership and organizational adaptability) that can influence (moderate) the relationship between AI adoption in businesses. AI capability development and deployments are resource-intensive, and an organization's leadership team must be aligned, prioritize AI projects, and form an AI governance team for developing ethical AI solutions. Industrial businesses should be adaptable, embrace changes, and develop a proper data-driven decision-making culture within the organization.

The study has several managerial implications, highlighting the critical AI capabilities of an industrial business. The managers should develop a comprehensive set of AI capabilities for their digital transformation projects. A robust data acquisition and management capability is a prerequisite for an AI-based business solution. Most of the time, data is in silos, and it is challenging to integrate real-time factory data with batch enterprise data. Managers must focus on integrating OT data with IT data to develop digital threads for their enterprise business processes for better proactive analysis, such as asset optimization, new product simulation, and proactive maintenance. Industrial businesses should focus on critical AI models for product development, service delivery, asset optimization, etc., and collect proper test data for model performance enhancement and testing. They should also focus on ethical AI development. Other than model development and deployment (technological) capabilities, good quality data can help in managerial decision-making. A robust cloud-based infrastructure is the foundation of AI-based projects. The IT managers should work with functional managers across the organization and plan for enterprise-wide AI projects such that they can share computing resources and AI models for different business applications. Skilled resources are critical for AI projects, and managers should hire critical resources from outside and train in-house resources for AI projects. Managers can leverage DDBMs to improve operational efficiency, increase customer-centricity, develop new ecosystem business models, and implement servitization. Managers should pay special attention to AI skills and leadership and enhance the organization's adaptability to AI projects; otherwise, they cannot realize the benefits of DDBM innovation.

The study has several limitations. First, we interviewed twenty-five elite informants from five high-technology companies, and based on their discussions, we developed these propositions. However, a large sample of companies from various industries should be included for validation. This can be a scope for future research. In this study, we have concentrated on industrial businesses; however, market segment-specific differences may exist and should be taken care of, and sector-specific differences in AI developments can be a part of a future study. Second, we have interacted with executives from large industrial companies (revenue over \$10B); however, there can be differences in small and medium (SMB) segments, and a future study should take care of those differences. Third, the study is an exploratory study of AI capabilities and their influence on DDBM innovation, and future longitudinal research can help to identify other factors that can influence DDBM innovation.

More and more industrial businesses are engaged in digital transformation initiatives and leveraging AI capabilities to develop new DDBMs. However, industrial managers are still debating the importance of AI in gaining competitive advantages. Researchers have been studying Big Data analytics and AI capabilities for a while; however, this paper presents a comprehensive set of AI capabilities for a firm and posits that data, technology, and foundation capabilities are critical for high-value, high-stake AI projects.

The study has also identified factors determining the AI-enabled value creation, delivery, and capture processes of DDBM innovation. Industrial managers can leverage multiple DDBM innovation types, such as operational efficiency, customer centricity, ecosystem-based platforms, and servitization, to redefine their current business models. Though AI is transforming our industrial businesses, strong leadership is essential for developing and implementing ethical AI practices. The managers should pay special attention to the biases and hallucinations of the models' outcomes. Adequate change management and governance procedures should be implemented for successful AI implementation.

CRedit authorship contribution statement

Swapan Ghosh: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jengtecman.2024.101851](https://doi.org/10.1016/j.jengtecman.2024.101851).

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